

Code-based Cryptography: Implementations and Cryptanalysis

Bernd Stelzl

Sebastian David Albu

Cryptanalysis ST 2024

Abstract

Coding theory as a field of mathematics predates its applications in cryptography. In the late 1950s, the work of R. W. Hamming first introduced coding theory as a field with the central question of transmitting data over a lossy symmetric channel while detecting and correcting resulting transmission errors.

Applications of coding theory in cryptography surfaced in the late 1970s. This paper will focus on the cryptosystem introduced in 1978 by McEliece due to its properties being representative of the promising opportunities coding theory offers for post-quantum cryptography, as well as its recent popularity as a finalist in the NIST PQC competition.

Keywords: Coding Theory · Error-Correcting Codes (Hamming) · McEliece · Public-Key-Encryption · Cryptanalysis · Post-Quantum Cryptography

Contents

1	Introduction	3
2	Background on Coding Theory	3
2.1	Algebraic Preliminaries	3
2.2	Error-correcting Codes	4
2.3	Syndrome Decoding	6
3	Applications of Coding Theory - McEliece Cryptosystem	6
3.1	Setup	6
3.2	Parameters and Hardness Assumption	7
4	Attacks and Cryptanalysis on McEliece	7
4.1	Adaptive Chosen Ciphertext Attack	7
4.2	Cryptanalysis with error patterns of the Hamming weight	9
4.3	Algebraic Cryptanalysis	10
5	Coding Theory and Post-Quantum Cryptography	12
6	Conclusion	12

1 Introduction

Coding theory has many advantages. The origins of coding theory date back to R. W. Hamming [Ham50]. He first introduced a code that can detect errors in the transmission using fundamental concepts of linear algebra, such as matrices, the distance between vectors, and the number of non-zero elements in a vector, commonly known as *Hamming weight*.

Coding Theory later became of interest to cryptographers, using the concepts of codes and their ability to maintain integrity and authenticity. McEliece [McE78] first introduced a public key encryption scheme that relies on coding theory.

Peter Shor [Sho94] introduced in 1994 a new algorithm that utilizes quantum computing and can exponentially speed up the discrete logarithm, as described in [McC90] and integer factorization, as described in [GS19], problems. Coding Theory is still popular amongst researchers since it is considered post-quantum secure. Shor's algorithm cannot break the hardness assumptions based on coding theory, as coding theory does not rely on the discrete logarithm problem or factoring.

This seminar paper will review the fundamentals of coding theory, some applications in cryptography, and attacks, giving an overview of the current state of coding theory in cryptography and the opportunities it brings.

Outline. In section 2, we introduce the mathematical background needed to understand the fundamentals behind coding theory. In section 3, we describe the public-key encryption scheme proposed by McEliece built upon these fundamentals. Following, we describe attacks and the approaches for cryptanalysis on the McEliece cryptosystem in section 4. In section 5, we discuss the current standing of code-based cryptography in the contemporary cryptography landscape, emphasize open research questions, and provide an outlook on future research. Finally, in section 6, we discuss our findings and give directions for future work.

2 Background on Coding Theory

It is imperative that we can depend on the accuracy of the information we receive, even when it is transmitted over inherently imperfect networks and prone to errors. This reliability is achieved through the principles of coding theory, which employs error-correcting codes to detect and rectify errors during transmission. For this section, we rely on the work of Luca Trevisan [Tre04].

2.1 Algebraic Preliminaries

Coding theory relies heavily on concepts from Linear and Abstract Algebra. The first step involves the selection of the alphabet $\mathbb{F}_q$.

Finite Field $\mathbb{F}_q$ A finite field $\mathbb{F}_q$ (also known as a *Galois field*) is a field with a finite number of elements, where q is a power of a prime number, i.e., $q = p^m$ for some prime p and positive integer m . In a finite field $\mathbb{F}_q$, the field operations (addition and multiplication) are defined and satisfy the usual properties of a field.

Let $\mathbb{F}_q^n$ denote a n -dimensional vector space over the finite field $\mathbb{F}_q$. A *linear code* C can now be defined.

Linear Codes For this seminar paper, we will only work with *linear codes*. A linear code is a subspace of $\mathbb{F}_q^n$. Specifically, with $n \in \mathbb{N}$ and a power of a prime q , we define $V(n, q) := \mathbb{F}_q^n$. A linear code $C \subset V(n, q)$ must satisfy the following properties:

1. $\forall \mathbf{u}, \mathbf{v} \in C : \mathbf{u} + \mathbf{v} \in C$
2. $\forall a \in \mathbb{F}_q \forall \mathbf{u} \in C : a\mathbf{u} \in C$

If C is a k -dimensional subspace of $V(n, q)$, then we also call C an $[n, k]$ -code or an $[n, k, d]$ -code, with d being the minimum distance of C , whereas d is also called the *Hamming distance*.

The elements of the code C are called codewords and are represented as n -dimensional vectors. The *generator matrix* G , which is a $k \times n$ (q -ary) matrix, can obtain those. Then it holds $\forall \mathbf{c} \in C, \forall m \in \mathbb{F}_q^k : \mathbf{c} = mG$.

2.2 Error-correcting Codes

The codeword $\mathbf{c}$ is now sent over an insecure channel, and errors are likely to occur. This is where error-correcting codes come into play. For this subsection, we rely on the work by MacWilliams and Sloane [MS77].

A vector $\mathbf{e}$ then denotes an error applied to a codeword $\mathbf{c}$. Arithmetically, we denote this as adding the error vector to a transmitted codeword. The received codeword $\mathbf{y}$ is then given by

$$\mathbf{y} = \mathbf{c} + \mathbf{e}.$$

If the Hamming weight of $\mathbf{e}$ is not too large, it is possible to recover still the original codeword $\mathbf{c}$, or at least parts of it. Here, *dual codes* are needed.

Dual Codes The dual code of a linear code C , typically denoted as $C^\perp$, consists of all vectors in $V(n, q)$ that are orthogonal to every codeword in C . In other words, $C^\perp$ is defined as follows:

$$C^\perp = \{\mathbf{w} \in V(n, q) \mid \mathbf{u} \cdot \mathbf{w} = 0 \quad \forall \mathbf{u} \in C\}$$

Here, $\mathbf{u} \cdot \mathbf{w}$ denotes the standard dot product in $V(n, q)$.

A generator matrix of the dual code $C^\perp$, usually denoted as H , is called the *parity check matrix* of the original code C . This is because it holds that

$$\mathbf{c}H^T = \mathbf{0} \quad \forall \mathbf{c} \in C.$$

It is also true that generating every codeword from C with this parity check matrix H is possible. We first need to prove that $(C^\perp)^\perp = C$.

Proof We want to show $(C^\perp)^\perp = C$

$$\begin{aligned} (C^\perp)^\perp &= \{\vec{v} \in V(n, q) \mid \vec{v} \cdot \vec{u} = 0 \quad \forall \vec{u} \in C^\perp\} \\ \supseteq: \text{ Let } \vec{v} \in C &\implies \forall \vec{u} \in C^\perp : \vec{v} \cdot \vec{u} = \vec{0} \implies \vec{v} \in (C^\perp)^\perp \\ \subseteq: \text{ Let } \vec{v} \in (C^\perp)^\perp &\implies \forall \vec{u} \in C : \vec{v} \cdot \vec{u} = \vec{0} \implies \vec{v} \in C \end{aligned}$$

It then also holds that

$$C = \left\{ \vec{x} \in V(n, q) : \vec{x} \cdot H^T = \vec{0} \right\} = \ker(H^T).$$

The main question remains on how these codes now can detect errors in $\mathbf{y}$ and how many errors can be detected by such a code. For this, we first need the *Hamming distance*, first introduced and named after R. W. Hamming [Ham50].

Hamming Distance and Minimal Distance The Hamming distance $d_H(\mathbf{x}, \mathbf{y})$ can be calculated using the Hamming weight w_H . It holds that

$$d_H(\mathbf{x}, \mathbf{y}) = w_H(\mathbf{x} - \mathbf{y}).$$

The minimal distance d between codewords in C can then be defined as the minimum Hamming weight, as the null vector $\vec{0}$ is always part of a linear code C .

With the minimal distance d of C , it is possible to determine the *error-correcting capability* t of C as

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

By virtue of the triangle inequality, it can be deduced that for every element y in $\mathbb{F}_q^n$, there exists at most one element c in C with a Hamming distance of at most t to it. Consequently, in theory, it becomes feasible to rectify up to t errors imposed on an element in C by identifying the nearest point within C . In practice, the decoding can be quite hard. An approach for efficient decoding is the *Syndrome decoding*.

2.3 Syndrome Decoding

The Syndrome Decoding problem was proven NP-complete in 1978 by Berlekamp, McEliece, and van Tilborg [BMV78]. For this subsection, we rely on their work and findings.

Let H be a parity check matrix of an $[n, k]$ -code C and $\vec{y} \in V(n, q)$. The vector $S(\vec{y}) := \vec{y} \cdot H^T$ is called the syndrome of $\vec{y}$. Thus, $S(\vec{y})$ is an $(n - k)$ -row vector.

If $\vec{h}_1, \dots, \vec{h}_{n-k}$ denote the rows of H , then $S(\vec{y}) = \vec{y} \cdot H^T = (\vec{y} \cdot \vec{h}_1, \dots, \vec{y} \cdot \vec{h}_{n-k})$. $S(\vec{y}) = 0$ holds if and only if $\vec{y}$ is orthogonal to each of the $\vec{h}_i$. In this case, $\vec{y} \in C$ (because H is the basis matrix of the orthogonal complement).

Lemma Two vectors $\vec{u}, \vec{v}$ lie in the same coset of a linear code C if and only if $S(\vec{u}) = S(\vec{v})$.

Proof It holds that $\vec{x} \cdot H^T = \vec{y} \cdot H^T$ if and only if $(\vec{x} - \vec{y}) \cdot H^T = \vec{0}$, thus $\vec{x} - \vec{y} \in C$. From group theory, we know that this is the case if and only if $\vec{x}$ and $\vec{y}$ lie in the same coset of C .

This establishes a connection between vectors lying in the same coset of a linear code and having the same syndrome. The proof demonstrates that if two vectors have the same syndrome, their difference lies in the code C , implying that they belong to the same coset. This insight is valuable for practical decoding purposes, as it allows for the calculation of syndromes for coset leaders, reduces the storage requirements and computational complexity involved in error correction processes, and allows for efficient decoding and error correction of transmitted codewords.

3 Applications of Coding Theory - McEliece Cryptosystem

Having covered the background needed for coding theory, it is time to demonstrate one representative cryptosystem built upon these ideas. The public key cryptosystem is based on the hardness of decoding and is proposed even to be hard enough when efficient Syndrome decoding is applied.

3.1 Setup

For the setup, we refer to the original publication by McEliece [McE78]. Let there be two parties called Alice and Bob. Let G be the generator matrix of a q -ary $[n, k, d]$ code C , constructed by Bob. Let $0 \leq t \leq e = \lfloor \frac{d-1}{2} \rfloor$ and let A_t be an effective decoding algorithm for C that can correct up to t errors. For cryptographic use, Bob creates a random, invertible q -ary $k \times k$ matrix S and a random $n \times n$ permutation matrix P . Bob's public key is $G' = SGP$ along with t , while S , G , and P remain secret. The matrix G' generates a code space C' equivalent to C but appears random, making decoding hard.

To encrypt a message $m \in F_q^k$ for Bob, Alice generates a random error vector $\vec{e}$ with $w(\vec{e}) \leq t$ and computes:

$$r = mG' + \vec{e} \quad (= mSGP + \vec{e}).$$

Upon receiving r , Bob computes:

$$rP^{-1} = (mS)G + \vec{e}P^{-1}.$$

Since $\vec{e}P^{-1}$ retains the weight of $\vec{e}$, Bob uses his decoding algorithm A_t to find mS (and $\vec{e}P^{-1}$) from rP^{-1} . Given that S is invertible, Bob can easily recover m , for example using Gaussian elimination.

3.2 Parameters and Hardness Assumption

For the McEliece cryptosystem, careful design decisions must be made. For this section, we refer to the publication by Luca Trevisan [Tre04] and McEliece [McE78]. In the original McEliece scheme, binary (i.e., $q = 2$) irreducible codes are used with parameters $n = 1024$, $k \approx 524$, and $t = 50$.

For the hardness assumptions to hold, the following parameters have been considered:

- With $k \approx \frac{n}{2} \geq 512$, syndrome decoding and exhaustive searches for the nearest codeword become infeasible.
- With $50 \leq t \leq 100$, techniques based on guessing/finding k (almost) error-free coordinates are less time-consuming than previous methods, but still infeasible.

These parameters have aged poorly, as stated by Faugère et al. [FOP+16] “Today, this is clearly beyond classical Gröbner bases algorithms”. Also stated in the paper is that McEliece, despite being post-quantum secure, has, up until the year 2016, never been applied to any big or significant real-life usages. The reason is believed to be the needed key size, not the known attacks.

4 Attacks and Cryptanalysis on McEliece

In this section, three different types of attacks will be used. In these attacks, we will mention *Goppa codes* and *Bose–Chaudhuri–Hocquenghem (BCH) codes*, which are some popular linear error-correcting codes. The Goppa codes are used in the McEliece cryptosystem, as explained in the publication by Valentijn [Val15]. The BCH codes are used, as in the original publication by Bose, Chaudhuri, and Hocquenghem [BR60].

4.1 Adaptive Chosen Ciphertext Attack

In the publication by Trevisan [Tre04], it has been proven that there exists an adaptive chosen ciphertext attack on the McEliece cryptosystem. The chosen ciphertext attack can be summarized as follows.

Maximal Error Property On input of a vector $y \in F_q^n$, the decoding algorithm A_t will return a codeword c in C at distance $\leq t$ to y if such a codeword exists. Otherwise, it will return an error message. Note that this property states that the decoding algorithm

A_t will not try to correct $t + 1$ errors. This is not possible in general if $t = \lfloor \frac{d-1}{2} \rfloor$. It is important to realize that if too many transmission errors have occurred, the received vector r may be at distance $\leq t$ from another codeword than the transmitted one. In this case, A_t will not return an error message. The probability that this occurs will be of importance in the analysis of the attack below.

Setup Assume the decoding algorithm A_t used in a McEliece cryptosystem has the maximal error property. Let r be the ciphertext Alice sent and Eve intercepted, where

$$r_A = mG_0 + e_A.$$

Eve performs the following steps:

Step 1: Increase the Number of Errors to Exactly t Eve increases the number of errors to the maximum by:

- Repeatedly changing a random coordinate in r and sending the altered codeword to Bob.
- Continuing this process until Bob returns an error message indicating that the message contains one error too many.
- Knowing that the previous message contained exactly t errors, Eve proceeds with this message r' to the next step.

Step 2: Determine Error-Free Coordinates With the message r' containing exactly t errors, Eve:

- Changes random coordinates in r' (excluding those modified in Step 1) and sends these to Bob.
- Identifies error-free coordinates when Bob returns an error message.
- Continues this process until enough error-free coordinates are identified for the next step.

Step 3: Determine the Plaintext Once enough error-free coordinates are known, Eve:

- Solves the matrix equation $r' = mG_0 + e'$ for the plaintext m . This step is crucial because knowing which coordinates are error-free allows Eve to accurately isolate the plaintext m from the received vector r' . The equation $r' = mG_0 + e'$ represents the relationship between the received vector, the plaintext, and the error vector. By solving this equation, Eve can determine the values of m .

- Uses Gaussian elimination on the columns corresponding to the known error-free coordinates to find m . Gaussian elimination is used here to solve the linear system of equations that arises from the matrix equation $r' = mG_0 + e'$. By focusing on the columns corresponding to the error-free coordinates, Eve can reduce the system to a simpler form, making it possible to solve for the plaintext m efficiently.

With this, Eve has completed an attack using adaptive chosen ciphertexts. While in theory, this attack is proven to be correct by the work of [Tre04], in practice, there has not been any real applied attack to this day. However, this is also because of the absence of the McEliece scheme in real-world applications.

Countermeasures Trevisan [Tre04] also has denoted some possible countermeasures to this attack in his publication. The following strategies are proposed to counter adaptive ciphertext attacks (CCA) on the McEliece cryptosystem: First, Bob can check for repeated messages to detect attacks. However, Eve can bypass this by adding a random codeword from the code C to each probe, maintaining the same error vector e and continuing the attack with minimal effort. Second, the error vector's weight can be fixed to t (or $t' \leq t$). If an error vector of different weight is encountered during decryption, an error message is returned regardless of successful decoding. Despite this, effective attacks remain possible: for non-binary codes, probing a coordinate twice with different values will show whether an error exists at that coordinate; for binary codes, changing any coordinate will always trigger an error message if error-free. By altering additional coordinates, one can identify the error-free and error-prone coordinates. Third, additional redundancy is introduced to prevent active eavesdroppers: Alice selects her plaintext m from F_q^{k-128} and a random error vector of weight $\leq t$. Bob publishes a cryptographic hash function h mapping bit strings to elements in F_q^{128} . Alice encrypts m as $r = (m || h(m || e))G_0 + e$. Bob decodes r' and checks the hash value of the decoded message and error vector. If verification fails, an error message is issued, revealing no information to Eve about e . The choice of 128 bits for the hash is arbitrary and can vary based on the security requirements. Note that this approach compromises the McEliece cryptosystem's inherent error-correcting capabilities, which seems unavoidable for this level of security.

4.2 Cryptanalysis with error patterns of the Hamming weight

In the publication from 1991 by Korzhik and Turkin [KT91], there are the first signs of cryptanalysis done on the McEliece public key cryptosystem.

First Approach: Finding Nonsingular Submatrices One approach is to find k coordinates where the matrix G' contains a nonsingular $k \times k$ submatrix and where the error vector e has only zeros in these positions. This makes decoding feasible by inverting the submatrix.

Second Approach: Equivalent Generator Matrices A second approach leverages the fact that G' might be equivalent to the generator matrix of another Goppa code, allowing the attacker to correct the error pattern e . However, the probability of this occurring is very small.

Third Approach: Iterative Optimization Algorithm The third and most notable approach involves using an iterative optimization algorithm. This algorithm guarantees the correction of error patterns with weight up to $\lfloor (d-1)/2 \rfloor$ for any linear code with minimum distance d . The process is as follows:

1. Embed binary vectors into the continuous vector space $\mathbb{R}^n$.
2. Reduce the decoding problem to finding extrema in $\mathbb{R}^n$ under code constraints.
3. Use an iterative procedure to find the desired extremum for error patterns with weight at most $\lfloor (d-1)/2 \rfloor$.

This approach does not contradict the NP-completeness of the general decoding problem because it applies only to words within a specific distance from a code word.

Application to (63, 26) Code To demonstrate this method, Korzhik, and Turkin applied it to a (63, 36) BCH code with minimum distance $d = 11$. They transformed the generator matrix G using random nonsingular and permutation matrices to get G' and tested the cryptanalysis with error patterns of Hamming weight 5. In 100 trials, the correct message was found every time, requiring about 10^6 operations, significantly less than the 5×10^6 operations for an exhaustive attack.

General Application For an arbitrary (n, k) code, this attack requires approximately $2^{0.113n}$ operations. Specifically, for a (1024, 654) BCH code with $d = 65$, the attack was demonstrated on an IBM PC during Eurocrypt '91 and back then took about 60 hours.

4.3 Algebraic Cryptanalysis

While the attacks presented above were able to reduce the complexity of breaking the cryptosystem by notable factors, the continued relevance of the scheme throughout the years has prompted research on more complex algebraic attacks [FOP+16; OTD10]. In this section, we will give an overview of an attack presented in 2016 by Faugère et al. [FOP+16] and an attack from 2010 presented by Otmani, Tillich, and Dallet [OTD10]. Both attacks respectively target attempts at reducing the size of the public keys used in the scheme in an attempt to move further towards practical applicability [FOP+16; OTD10]. Due to both attacks assuming an understanding of contemporary mechanisms to reduce the size of the public-key, thus exceeding the scope of this paper, we will focus mainly on giving a general intuition on the attacks and their respective complexity.

Faugère et al. [FOP+16] One approach for reducing public-key size was presented by Berger et al. [BCGO09]. This approach uses the algebraic properties of certain classes of aforementioned, *Goppa codes* resulting in more compact generator and parity-check matrices and, therefore, smaller keys. The resulting matrices are usually *quasi-cyclic*, *quasi-dyadic*, or *quasi-monoidic*. Intuitively, a *cyclic* code denotes a code such that applying a linear shift operation onto a codeword always results in another word within the code. The distinction follows algebraically from viewing this property of *cyclic* codes as an ideal within a quotient ring, where *quasi-cyclic* generalize this to constituting modules instead. Intuition for *quasi-monoidic* and *quasi-dyadic* codes follows analogously [FOP+16].

First attacks showing the diminishing security level resulting from these reductions have been shown shortly after the initial proposal but targeted mostly instantiations with specific properties [FOP+16]. Meanwhile, Faugère et al. [FOP+16] improves attack complexity and generalizes the attack toward a larger class of instances resulting from this technique [FOP+16].

The applied strategy is based on the "Structural Elimination" technique, essentially allowing the reduction of the number of variables within an algebraic description of the cryptosystem, thus reducing the complexity of former attacks [FOP+16].

Experimental results show that depending on a specific problem (Encryption, Signature, ...) and if the attack is applicable, it leads to swift key recovery [FOP+16]. In conclusion, this attack showcased inherent weaknesses in reducing public key size, removing a possible avenue toward practical applications of the cryptosystem.

Otmani, Tillich, and Dallot [OTD10] Similarly to the attack presented by Faugère et al. [FOP+16], this attack also targets key size reductions based on *quasi-cyclic* codes. The paper instead of providing a generalized approach focuses on variants using specific subcodes of *BCH*- and *LDPC*-codes [OTD10]. Especially the latter being of significant interest due to such constructions specifically intended to mitigate prior cryptanalysis performed on the cryptosystem [OTD10]. Similarly to the attack presented before, this attack leverages properties stemming from the *quasi-cyclic* nature of the code and, adding certain linear transformations employed in these codes, attempts to find pairs of polynomials of low-weight to recover the parity-check matrix (secret-key) [OTD10]. While the attack is inherently probabilistic, it comes at a respectable success rate, resulting in an overall complexity of $\mathcal{O}(n^3)$ depending on the code's length n .

5 Coding Theory and Post-Quantum Cryptography

What makes the McEliece cryptosystem especially notable is the properties it is believed to have despite being presented a significant time ago. As a finalist among the NIST PQC standardization process, it stands out among the other, overall very recent, candidates [WLL+22]. Given that, as shown in the prior section, attempts at reducing the public key size have resulted in powerful attacks, key size remains an open problem to be considered in practical applications.

In terms of security, the basic variant of the scheme still offers promising properties in PQC, as no polynomial-time quantum algorithm has been found so far, and the incorporation of randomness within the schemes could open the door to further applications in the area of privacy [WLL+22].

Another challenge is posed by the minimal amount of coding-theory schemes aside from McEliece. While promising schemes such as the Niederreiter algorithm exist, this still raises questions when compared with other cryptographic problems studied in many implementations [WLL+22]. In contrast to this stands the fact that schemes based on coding theory show similarities to those based on lattices in many aspects, allowing for partial security analysis to be built upon reductions between the two mathematical objects [WLL+22].

A larger selection of codes to build a cryptosystem would create more resilience should efficient quantum algorithms on specific sets of codes be found.

6 Conclusion

Code-based cryptography, with the McEliece cryptosystem as an important representative, showcases many properties of interest. Aside from security claims that still bear weight under contemporary requirements, the system allows for constructing various schemes in the toolkit necessary for modern cryptography.

On the other hand, these ideas are currently held back from practical applications due to large key sizes, with prior attempts at reductions going hand in hand with significant security implications. Therefore, important research questions are finding secure ways to reduce the key size and devising novel cryptosystems built upon coding theory, possibly leveraging the promising properties shown in a more practical setting. To conclude, while code-based cryptography may not be at the center of attention in post-quantum efforts, it has produced a promising candidate that is giving direction toward further research, with the opportunity of creating practical schemes in the future.

References

- [BCGO09] Thierry P Berger, Pierre-Louis Cayrel, Philippe Gaborit, and Ayoub Otmani. “Reducing key length of the McEliece cryptosystem”. In: *International Conference on Cryptology in Africa*. Springer. 2009, pp. 77–97.
- [BMV78] Elwyn Berlekamp, Robert McEliece, and Henk Van Tilborg. “On the inherent intractability of certain coding problems (corresp.)”. In: *IEEE Transactions on Information theory* 24.3 (1978), pp. 384–386.
- [BR60] Raj Chandra Bose and Dwijendra K Ray-Chaudhuri. “On a class of error correcting binary group codes”. In: *Information and control* 3.1 (1960), pp. 68–79.
- [FOP+16] Jean-Charles Faugère, Ayoub Otmani, Ludovic Perret, Frédéric De Portzamparc, and Jean-Pierre Tillich. “Structural cryptanalysis of McEliece schemes with compact keys”. In: *Designs, codes and Cryptography* 79.1 (2016), pp. 87–112.
- [GS19] Pinkimani Goswami and Madan Singh. “Integer Factorization Problem”. In: May 2019, pp. 59–77. ISBN: 9781351021708. DOI: 10.1201/9781351021708-5.
- [Ham50] R. W. Hamming. “Error detecting and error correcting codes”. In: *The Bell System Technical Journal* 29.2 (1950), pp. 147–160. DOI: 10.1002/j.1538-7305.1950.tb00463.x.
- [KT91] Valery I Korzhik and Andrey I Turkin. “Cryptanalysis of McEliece’s public-key cryptosystem”. In: *Workshop on the Theory and Application of Cryptographic Techniques*. Springer. 1991, pp. 68–70.
- [McC90] Kevin S McCurley. “The discrete logarithm problem”. In: *Proc. of Symp. in Applied Math.* Vol. 42. USA. 1990, pp. 49–74.
- [McE78] Robert J McEliece. “A public-key cryptosystem based on algebraic”. In: *Coding Thv* 4244 (1978), pp. 114–116.
- [MS77] Florence Jessie MacWilliams and Neil James Alexander Sloane. *The theory of error-correcting codes*. Vol. 16. Elsevier, 1977.
- [OTD10] Ayoub Otmani, Jean-Pierre Tillich, and Léonard Dallot. “Cryptanalysis of two McEliece cryptosystems based on quasi-cyclic codes”. In: *Mathematics in Computer Science* 3 (2010), pp. 129–140.
- [Sho94] Peter W. Shor. “Algorithms for quantum computation: discrete logarithms and factoring”. In: *Proceedings 35th Annual Symposium on Foundations of Computer Science* (1994). Accessed: 2024-04-28, pp. 124–134. URL: <https://api.semanticscholar.org/CorpusID:15291489>.
- [Tre04] Luca Trevisan. “Some applications of coding theory in computational complexity”. In: *arXiv preprint cs/0409044* (2004).

- [Val15] Ashley Valentijn. “Goppa codes and their use in the McEliece cryptosystems”. In: (2015).
- [WLL+22] Jiabo Wang, Ling Liu, Shanxiang Lyu, Zheng Wang, Mengfan Zheng, Fuchun Lin, Zhao Chen, Liuguo Yin, Xiaofu Wu, and Cong Ling. “Quantum-safe cryptography: crossroads of coding theory and cryptography”. In: *Science China Information Sciences* 65.1 (2022), p. 111301.